Notes: Heblich, Redding, and Sturm (QJE, 2020)

The Making of the Modern Metropolis: Evidence from London

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1 Overview

1.1 Contribution

The paper quantifies how a large commuting-technology shock—the construction of London's overground and underground railways—reshaped the internal spatial equilibrium of the city (residence vs. workplace locations) and created sizable aggregate welfare/value gains, using a combination of reduced-form evidence and a quantitative spatial equilibrium model.

1.2 Core mechanism in words

Lower commuting costs relax the "time-to-center" constraint. That allows:

- **Spatial separation:** people live farther away while keeping access to high-wage/high-productivity jobs.
- **Functional specialization:** central locations become relatively more workplace-intensive; outer locations become relatively more residential.
- **Land-value capitalization:** because access improves, the willingness to pay for residential and commercial floorspace increases and is capitalized into rateable values.
- **Amplification:** if there are agglomeration forces (productivity and/or amenities rising with density), the reallocation of activity across space magnifies total gains.

1.3 What the paper delivers

1. A clean **commuting-cost elasticity** from a gravity-style commuting equation (Table I).
2. A **two-sided spatial equilibrium system** with residence/workplace choices and land-market clearing (Equations (3)–(16)).
3. An **exact-hat-algebra** method that backs out wage changes and commuting patterns from observed changes in residents and land values plus commuting-cost changes (Lemma 1; Equations (17)–(21)).
4. A way to recover **changes in composite productivity and amenities** and estimate **agglomeration elasticities** (Equations (30)–(31); Table II).
5. Welfare/value counterfactuals for transport-network removal, including the role of floor-supply elasticity and agglomeration (Table III; Figure IX).

2 Data, geography, and the empirical setting

2.1 Geographic units and outcomes

- The city is partitioned into fine spatial units (parishes/boroughs) inside **Greater London**.
- Key outcomes:

1. **Night population** (residents) by location $R_{n,t}$,
2. **Workplace employment** by location $L_{i,t}$,
3. **Rateable values** $Q_{n,t}$ (a land-value proxy that aggregates residential and commercial floorspace payments),
4. **Commuting flows** $L_{ni,t}$ in 1921 (residence n to workplace i),
5. **Railway network** over time, translated into travel-time matrices.

2.2 A key set of stylized facts (what must be matched)

Figures II–IV document the defining transformation:

- **City of London resident population collapses after mid-19th century**, while Greater London continues to expand (Figure II).
- **Day population rises as night population falls** in the central City, while the City's share of rateable values does not collapse (Figure III). This points to a shift toward workplace activity and land-value capitalization.
- **Travel-time indices fall** over time (railway expansion), coinciding with strong changes in residence and workplace distributions (Figure IV).

3 Reduced-form "railway as shock" evidence

This section describes the paper's reduced-form empirical designs that motivate the structural interpretation.

3.1 Event-study design around station arrival

For parish j in census year t (1801–1901), define the treatment-year index

$$\tau_{jt} \equiv t - \max\{t' \leq t : \text{parish } j \text{ has no railway station in census year } t'\}.$$

Thus, $\tau_{jt} > 0$ are post-treatment periods; $\tau_{jt} = 0$ is the last census year without a station.

The baseline event-study regression (paper Equation (1)) is:

$$\log(\text{Pop}_{jt}) = \alpha_j + \delta_t + \sum_{\tau=-\overline{\tau}}^{\overline{\tau}} \beta_\tau \, \mathbf{1}\{\tau_{jt} = \tau\} + \gamma_j t + \varepsilon_{jt}, \tag{1}$$

with parish fixed effects α_j, census-year fixed effects δ_t, and parish-specific linear trends $\gamma_j t$.

Interpretation.

- The coefficients β_τ trace the evolution of population around the first appearance of a station.
- Validity requires that (conditional on FE and trends) station arrival is not systematically timed to unobserved population shocks.

3.2 Central vs. outer heterogeneity

To allow differential effects in central areas, the paper estimates (paper Equation (2)):

$$\begin{aligned}\log(\text{Pop}_{jt}) = \alpha_j + \delta_t + \sum_\tau \beta_\tau \mathbf{1}\{\tau_{jt} = \tau\} + \sum_\tau \gamma_\tau \mathbf{1}\{\tau_{jt} = \tau\} \times \mathbf{1}\{j \in \text{Central}\} \\ + \sum_\tau \zeta_\tau \mathbf{1}\{\tau_{jt} = \tau\} \times TT_{jt} + \gamma_j t + \varepsilon_{jt}.\end{aligned} \tag{2}$$

Interpretation.

- β_τ is the event-study effect for non-central (outer) parishes.
- γ_τ is the *difference* in event-study effects between central and outer parishes.
- The travel-time interaction terms help address the idea that central vs outer differences might operate through heterogeneous time-to-center changes.

3.3 Reduced-form takeaway from Figure V

Figure V plots these estimated event-study coefficients:

- Effects are **positive post-arrival** for Greater London overall.
- Effects are **stronger for outer parishes** and much weaker (or negative relative to outer) for central parishes, consistent with suburbanization enabled by commuting.
- Pre-trends are limited once parish FE and trends are included (important for credibility).

4 Quantitative spatial equilibrium model

4.1 Spatial indices and accounting identities

Let $n \in \mathcal{N}$ denote a **residential** location and $i \in \mathcal{N}$ denote a **workplace** location (in practice, the same set of locations can play both roles).

Define:

- Residents (night population) in n: $R_{n,t}$.
- Workplace employment in i: $L_{i,t}$.
- Commuters from n to i: $L_{ni,t}$.
- Total Greater London population/employment: $L_{\mathcal{N},t} = \sum_n R_{n,t} = \sum_i L_{i,t}$.
- **Commuting probabilities:**

$$\lambda_{ni,t} \equiv \frac{L_{ni,t}}{L_{\mathcal{N},t}}, \qquad \lambda^R_{n,t} \equiv \frac{R_{n,t}}{L_{\mathcal{N},t}}, \qquad \lambda^L_{i,t} \equiv \frac{L_{i,t}}{L_{\mathcal{N},t}}.$$

- Conditional commuting probabilities (workplace choice conditional on residence):

$$\lambda^R_{ni|n,t} \equiv \frac{L_{ni,t}}{R_{n,t}}.$$

4.2 Preferences and indirect utility (paper Equation (3))

Workers have Cobb–Douglas preferences over a composite consumption good (price index $P_{n,t}$) and residential floorspace (price $Q_{n,t}$):

$$U_{ni,t}(\omega) = \frac{B_{ni,t}\, b_{ni,t}(\omega)\, w_{i,t}}{\kappa_{ni,t}\, P^{\alpha}_{n,t}\, Q^{1-\alpha}_{n,t}}, \qquad 0 < \alpha < 1, \tag{3}$$

where:

- $w_{i,t}$ is the wage paid at workplace i,
- $\kappa_{ni,t} \geq 1$ is an iceberg **commuting cost** from n to i,
- $B_{ni,t}$ is a **common** (deterministic) bilateral amenity component,
- $b_{ni,t}(\omega)$ is an **idiosyncratic** amenity/compatibility term for worker type ω.

4.3 Frechét idiosyncratic shocks (paper Equation (4))

Assume

$$\Pr(b_{ni,t}(\omega) \leq b) = G(b) = \exp\left(-b^{-\theta}\right), \qquad \theta > 1. \tag{4}$$

Why this matters. Frechét shocks deliver closed-form choice probabilities that look like a gravity equation: systematic utility terms raised to the power θ.

4.4 Decomposing bilateral amenities (paper Equation (5))

$$B_{ni,t} = B^R_{n,t}\, B^L_{i,t}\, B^I_{ni,t}, \tag{5}$$

where B^R is residence-specific, B^L is workplace-specific, and B^I is pair-specific.

4.5 Choice probabilities (paper Equation (6))

Define the *systematic* component in (3):

$$\tilde{U}_{ni,t} \equiv \frac{B_{ni,t}\, w_{i,t}}{\kappa_{ni,t}\, P^{\alpha}_{n,t}\, Q^{1-\alpha}_{n,t}}.$$

Under Frechét shocks (4), the probability that a worker chooses (n, i) among all residence–workplace pairs in London is:

$$\lambda_{ni,t} = \frac{\tilde{U}^{\theta}_{ni,t}}{\sum_{k\in\mathcal{N}} \sum_{j\in\mathcal{N}} \tilde{U}^{\theta}_{kj,t}}. \tag{6}$$

Derivation sketch. With Frechét heterogeneity, $\max_{(n,i)}\{\tilde{U}_{ni}b_{ni}\}$ has an extreme-value structure. The probability that (n,i) delivers the highest value equals its "attractiveness" $\tilde{U}_{ni}^{\theta}$ divided by the sum across all pairs.

4.6 Residence and workplace shares (paper Equations (7)–(8))

Summing (6) yields:

$$\lambda_{n,t}^{R} = \sum_{i\in\mathcal{N}} \lambda_{ni,t}, \qquad \lambda_{i,t}^{L} = \sum_{n\in\mathcal{N}} \lambda_{ni,t}. \tag{7}$$

Conditional workplace choice given residence n simplifies because $P_{n,t}$ and $Q_{n,t}$ cancel within a given n:

$$\lambda_{ni|n,t}^{R} = \frac{\left(\dfrac{B_{ni,t}\,w_{i,t}}{\kappa_{ni,t}}\right)^{\theta}}{\sum_{j\in\mathcal{N}}\left(\dfrac{B_{nj,t}\,w_{j,t}}{\kappa_{nj,t}}\right)^{\theta}}. \tag{8}$$

Define average (expected) per-capita income by residence:

$$v_{n,t} \equiv \sum_{i\in\mathcal{N}} \lambda_{ni|n,t}^{R}\, w_{i,t}. \tag{9}$$

4.7 Production and factor payments (paper Equation (10))

Firms in workplace location i produce output $Y_{i,t}$ with Cobb–Douglas technology in labor $L_{i,t}$, commercial floorspace $H_{i,t}^{L}$, and an outside input (e.g. machinery) $M_{i,t}$:

$$Y_{i,t} = A_{i,t}\, L_{i,t}^{\beta_L}\,(H_{i,t}^{L})^{\beta_H}\, M_{i,t}^{\beta_M}, \qquad \beta_L+\beta_H+\beta_M = 1. \tag{10}$$

Under perfect competition and zero profits,

$$w_{i,t}L_{i,t} = \beta_L Y_{i,t}, \quad q_{i,t}H_{i,t}^{L} = \beta_H Y_{i,t}, \quad r_t M_{i,t} = \beta_M Y_{i,t}. \tag{11}$$

4.8 Commercial vs. residential floorspace prices (paper Equations (11)–(12))

Commercial floorspace in i can have a wedge relative to residential floorspace:

$$q_{i,t} = \xi_{i,t}\, Q_{i,t}. \tag{12}$$

Combining (11) implies:

$$q_{i,t}H_{i,t}^{L} = \frac{\beta_H}{\beta_L}\, w_{i,t}L_{i,t}. \tag{13}$$

4.9 Commuter market clearing (paper Equations (13)–(15))

Workplace employment equals the sum of commuters arriving from all residences:

$$L_{i,t} = \sum_{n\in\mathcal{N}} \lambda_{ni|n,t}^{R}\, R_{n,t}. \tag{14}$$

Average income by residence is (9).

4.10 Land-market clearing and rateable values (paper Equation (16))

Rateable values summarize the total payments for residential and commercial floorspace in location n:

$$Q_{n,t} = (1-\alpha)\, v_{n,t}\, R_{n,t} + \frac{\beta_H}{\beta_L}\, w_{n,t}\, L_{n,t}. \tag{15}$$

Interpretation. Rateable values are a sufficient statistic for local land-market "pressure" coming from: (i) residents (via housing demand) and (ii) firms (via commercial floorspace demand).

5 Exact hat algebra: identifying wage and commuting changes from $(\hat{R}, \hat{Q}, \hat{\kappa})$

This is the key methodological bridge: the paper shows that with observed changes in residents and land values plus inferred changes in commuting costs, one can solve for wage changes and then generate counterfactual spatial allocations.

5.1 Hat notation

Fix a baseline year t (the paper's quantitative baseline is $t = 1921$) and another year $\tau \neq t$. For any variable $x_{n,t}$ define the relative change:

$$\hat{x}_{n,t} \equiv \frac{x_{n,\tau}}{x_{n,t}}.$$

Then $x_{n,\tau} = \hat{x}_{n,t}\, x_{n,t}$.

5.2 Rewriting land-market clearing in hats (paper Equation (17))

Starting from (15) in year τ and substituting $x_{n,\tau} = \hat{x}_{n,t} x_{n,t}$ yields:

$$\hat{Q}_{n,t} Q_{n,t} = (1-\alpha)\, \hat{v}_{n,t} v_{n,t}\, \hat{R}_{n,t} R_{n,t} + \frac{\beta_H}{\beta_L}\, \hat{w}_{n,t} w_{n,t}\, \hat{L}_{n,t} L_{n,t}. \tag{16}$$

5.3 Hatted commuter-market objects (paper Equations (18)–(19))

Using commuter market clearing and the conditional commuting probabilities, the paper expresses relative changes in workplace employment and residence income as functions of wage changes and commuting-cost changes:

$$\hat{L}_{i,t} L_{i,t} = \sum_{n \in \mathcal{N}} \left(\frac{\lambda^R_{ni|n,t}\, \hat{w}^{\theta}_{i,t}\, \hat{\kappa}^{-\theta}_{ni,t}}{\sum_{j \in \mathcal{N}} \lambda^R_{nj|n,t}\, \hat{w}^{\theta}_{j,t}\, \hat{\kappa}^{-\theta}_{nj,t}} \right) \hat{R}_{n,t} R_{n,t}, \tag{17}$$

$$\hat{v}_{n,t} v_{n,t} = \sum_{i \in \mathcal{N}} \left(\frac{\lambda^R_{ni|n,t}\, \hat{w}^{\theta}_{i,t}\, \hat{\kappa}^{-\theta}_{ni,t}}{\sum_{j \in \mathcal{N}} \lambda^R_{nj|n,t}\, \hat{w}^{\theta}_{j,t}\, \hat{\kappa}^{-\theta}_{nj,t}} \right) \hat{w}_{i,t} w_{i,t}. \tag{18}$$

Key modeling restriction used here. The paper assumes workplace and bilateral amenity components are constant over time ($\hat{B}^L_{i,t} = 1$, $\hat{B}^I_{ni,t} = 1$) so that amenities do not enter (17)–(18).

5.4 Solving for wages (paper Equation (20) and Lemma 1)

Substitute (17)–(18) into (16) for each n. This yields $|\mathcal{N}|$ equations in the unknown wage-change vector $\{\hat{w}_{n,t}\}_{n\in\mathcal{N}}$.

Lemma (existence/uniqueness). Given observed baseline objects $(\lambda^R_{ni|n,t}, L_{n,t}, R_{n,t}, Q_{n,t}, v_{n,t}, w_{n,t})$, observed changes $(\hat{R}_{n,t}, \hat{Q}_{n,t})$, parameters $(\alpha, \beta_L, \beta_H, \theta)$, and commuting-cost changes $\hat{\kappa}^{-\theta}_{ni,t}$, the system has a unique solution for $\{\hat{w}_{n,t}\}$ (paper Lemma 1).

5.5 Recovering commuting flows (paper Equation (21))

Once $\hat{w}$ is solved, the relative change in commuting flows follows from the conditional probabilities:

$$\hat{L}_{ni,t} L_{ni,t} = \left(\frac{\lambda^R_{ni|n,t}\, \hat{w}^{\theta}_{i,t}\, \hat{\kappa}^{-\theta}_{ni,t}}{\sum_{j\in\mathcal{N}} \lambda^R_{nj|n,t}\, \hat{w}^{\theta}_{j,t}\, \hat{\kappa}^{-\theta}_{nj,t}} \right) \hat{R}_{n,t} R_{n,t}. \tag{19}$$

Economic interpretation. This block says: *given* commuting-cost changes and observed changes in where people live and how much land is worth, the spatial equilibrium implies a unique implied change in workplace wages, and therefore a unique implied reorganization of commuting flows and workplace employment.

6 Estimating commuting costs from travel time (Table I and related figure evidence)

6.1 Linking commuting costs to railway travel time (paper Equation (22))

Let $d^W_{ni,t}$ be the **rail travel time** between n and i in year t (constructed from the network). The paper models commuting costs as a function of travel time, up to unobserved components:

$$-\theta \log \kappa_{ni,t} = -\theta\phi \log d^W_{ni,t} + u^R_{n,t} + u^L_{i,t} + u^I_{ni,t}. \tag{20}$$

Using (8), this implies a gravity regression:

$$\log \lambda^R_{ni|n,t} = -\theta\phi \log d^W_{ni,t} + \text{residence FE} + \text{workplace FE} + \text{bilateral residual}.$$

6.2 Table I: what is being estimated?

Table I uses 1921 commuting flows to estimate the sensitivity of commuting probabilities to travel time:

$$\log \lambda^R_{ni|n,1921} = \gamma \log d^W_{ni,1921} + \text{FE} + \varepsilon_{ni}, \quad \gamma = -\theta\phi.$$

Because the network can be endogenous to economic activity, the paper instruments $\log d^W$ with straight-line travel time $\log d^S$.

6.3 Table I: interpretation of the coefficients

- **OLS estimate:** $\gamma \approx -4.899$ (s.e. 0.062).
- **IV estimate:** $\gamma \approx -5.203$ (s.e. 0.069).
- **First stage:** $\log d^W$ loads positively on $\log d^S$ (coefficient ≈ 0.429), with a very large F-statistic.

Economic meaning.

- The magnitude around -5 implies commuting probability is *highly elastic* to travel time.
- A doubling of travel time reduces predicted commuting probability by roughly $2^{-5} \approx 1/32$ (holding FE fixed).
- This is exactly the kind of elasticity needed for railways to meaningfully reshape the internal spatial equilibrium.

7 Recovering productivity and amenities; estimating agglomeration (Table II)

7.1 Recovering changes in composite productivity

The model yields a composite production fundamental $A_{n,t}^T$ that is recoverable from equilibrium conditions once wages, employment, and land values are pinned down. The paper focuses on changes, using the exact-hat approach.

7.2 Recovering changes in composite amenities (paper Equation (29))

Using residence shares and the structure of (6), the paper derives an expression that backs out changes in composite amenities $\hat{B}_{n,t}$:

$$\hat{\lambda}_{n,t}^R \lambda_{n,t}^R = \frac{\lambda_{n,t}^R \, \hat{B}_{n,t}^{\theta} \, \hat{Q}_{n,t}^{-\theta(1-\alpha)} \, R_{n,t}^{MA}}{\sum_{k \in \mathcal{N}} \lambda_{k,t}^R \, \hat{B}_{k,t}^{\theta} \, \hat{Q}_{k,t}^{-\theta(1-\alpha)} \, R_{k,t}^{MA}}, \tag{21}$$

where $R_{n,t}^{MA}$ is a residence "commuting market access" term that depends on wage and commuting-cost changes (it is computed from $\hat{w}$ and $\hat{\kappa}$). Equation (21) identifies $\hat{B}_{n,t}$ up to a normalization.

7.3 Agglomeration in production and residence (paper Equations (30)–(31))

The paper allows density-dependent externalities:

- **Production agglomeration:** productivity increases with workplace employment density,

- **Residential agglomeration:** amenities increase with residential density.

Let K_n denote land area. Between baseline $\tau = 1831$ and $t = 1921$:

$$\log \hat{A}^T_{n,t} = \log \hat{a}_{n,t} + \eta^L \log \hat{L}_{n,t} + \log \hat{A}^T_t, \tag{22}$$
$$\log \hat{B}^R_{n,t} = \log \hat{b}_{n,t} + \eta^R \log \hat{R}_{n,t} + \log \hat{B}^R_t. \tag{23}$$

Here η^L and η^R are agglomeration elasticities, while $\hat{a}_{n,t}$ and $\hat{b}_{n,t}$ are residual (exogenous) fundamental changes.

7.4 Table II: interpretation

Table II estimates (22)–(23) using OLS and IV:

- **OLS:** $\eta^L \approx 0.148$ and $\eta^R \approx 0.248$ (both highly significant).
- **IV:** $\eta^L \approx 0.086$ and $\eta^R \approx 0.172$ (still significant, smaller magnitudes).

How to read this.

- OLS likely overstates agglomeration because density changes are endogenous to unobserved fundamentals.
- IV reduces the estimates, consistent with sorting/endogeneity bias in OLS.
- Both production and residential agglomeration are economically meaningful at these magnitudes: they act as multipliers in counterfactual transport experiments.

8 Counterfactuals: removing the railway network (Table III and Figure IX)

8.1 Conceptual counterfactual

Hold recovered *fundamentals* fixed (productivity and amenities net of agglomeration), and change commuting costs by altering the transport network:

$$\kappa^{CF}_{ni,t} \leftarrow \text{commuting costs implied by a counterfactual network.}$$

Then solve the spatial equilibrium system for wages, commuting flows, employment, residents, and land values.

8.2 Why assumptions matter

Two key assumptions govern magnitudes:

- **Floorspace supply elasticity** μ (how easily the city can add floorspace when land values rise).

- **Agglomeration elasticities** (η^L, η^R) (whether density changes feed back into productivity/amenities).

8.3 Table III: interpretation panel-by-panel

Table III reports welfare/value impacts from removing:

1. Panel A: **all railway lines** (overground + underground),
2. Panel B: **underground lines only**,
3. Panel C: **lines built during 1911–1921**.

How to read the columns. Each panel reports:

- **Economic impact:** change in rateable values (millions of pounds in 1921).
- **NPV at 3% and 5% discount rates:** present value of the rateable-value change.
- **Construction costs:** cost of the removed lines.
- **Impact / cost ratio:** a benefit-cost style statistic.

Panel A (all lines): headline message. Removing the entire network generates large losses, and the losses increase when you allow (i) elastic floorspace supply and (ii) agglomeration:

- With $\mu = 0$ and no agglomeration: economic impact ≈ -8.24.
- With $\mu = 1.83$ and no agglomeration: impact ≈ -14.35.
- With $\mu = 1.83$ and agglomeration $(\eta^L, \eta^R) = (0.086, 0.172)$: impact ≈ -35.07 and the impact/cost ratio rises to about 17.7 (NPV at 3%).

Panel B (underground only): interpretation. Even the underground alone accounts for a large share of the value creation: losses range from about -5.59 (no agglomeration, $\mu = 0$) to about -20.31 (with agglomeration, $\mu = 1.83$), with a sizable impact/cost ratio.

Panel C (1911–1921 lines): interpretation. The most recent pre-1921 expansions still generate nontrivial welfare gains: economic impacts are smaller in level but still have meaningful benefit/cost ratios.

8.4 Figure IX: what the counterfactual allocations look like

Figure IX visualizes how the city's *spatial organization* changes under railway removal:

- Commuting becomes more localized; long-distance commutes shrink.
- Central workplace concentration weakens; residence becomes less suburbanized.
- These spatial reallocations are the channel through which land values (and thus welfare) change.

9 Interpretation - figures and tables

This section is meant to be read side-by-side with the reproduced pages in the appendix.

9.1 Figure I: Administrative boundaries and the railway network over time

What it shows. Panels map Greater London, the historical County of London (LCC), the City of London, and overlay overground/underground lines across time.

How to interpret. The figure is the empirical "treatment map": it makes clear that the network expansion (especially into outer areas) is spatially heterogeneous, which is necessary for identification (cross-location variation in travel-time shocks).

9.2 Figure II: London population indices over time

What it shows. Time-series indices for City of London vs Greater London (1801–1921).

Key message. Greater London expands dramatically, but the City's *resident* population falls sharply after the mid-19th century. Any model must explain: strong aggregate growth *and* central depopulation.

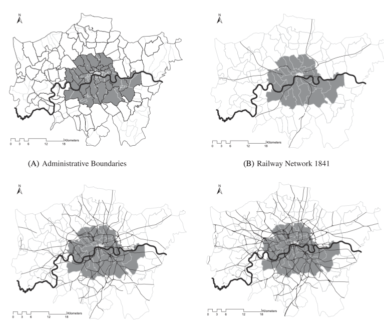


FIGURE I
Administrative Boundaries and the Railway Network over Time

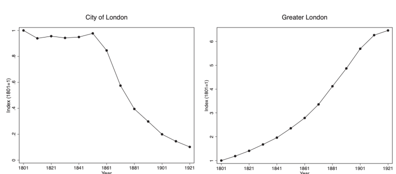


FIGURE II
Population Indexes over Time (City of London and Greater London, 1801 = 1)
Indexes of residence (night) population from the population censuses over time.

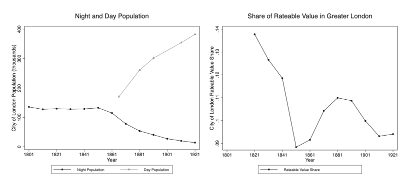


FIGURE III
City of London Day and Night Population and Rateable Value Share over Time

9.3 Figure III: City day vs night population and rateable-value share

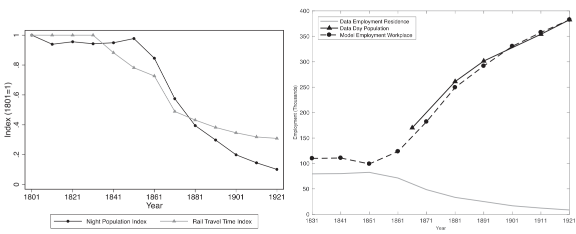
What it shows. Two linked facts for the City of London: (i) night (resident) population collapses while day population rises, and (ii) the City's share of rateable values does not collapse in the same way.

Intuition. The City becomes increasingly specialized as a workplace/office center. High willingness-to-pay for central commercial floorspace keeps land values high even when residents leave.

9.4 Figure IV: Travel time, night population, and employment (observed vs model)

What it shows. A time-series of (a) weighted travel-time indices (falling with rail expansion) and (b) the evolution of employment by workplace and residence, with a model-implied counterpart.

Key message. Once you feed in commuting-cost changes, the model can replicate key movements in workplace employment. This is the "credibility check" that the commuting-cost channel has quantitative power.



(A) Night Population and Rail Travel Time Indices (B) Employment by Residence and Workplace

FIGURE IV

Night Population, Weighted Average Travel Time, Employment by Residence, and Employment by Workplace in the City of London

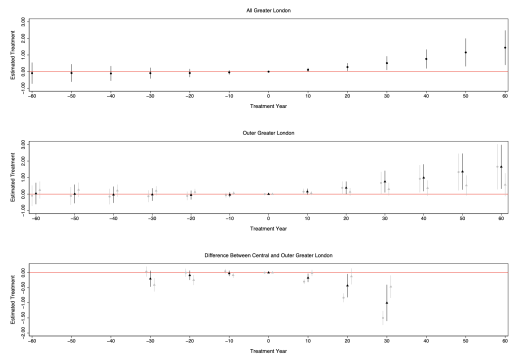


FIGURE V

Event-Study Treatment Effects for the Arrival of an Overground or Underground Railway Station for Greater London Parishes from 1801 to 1901

9.5 Figure V: Event-study treatment effects around station arrival

What it shows. Event-study coefficients for station arrival effects on log population for: (i) all Greater London, (ii) outer Greater London, (iii) difference between central and outer.

Key interpretation. Post-arrival growth is stronger in outer areas and weaker in central areas. This supports the structural claim: railways facilitate suburban residence while maintaining central employment access.

9.6 Table I: Estimation of commuting costs

What it estimates. A gravity-style commuting equation linking commuting probabilities to travel time.

Main conclusion. The elasticity around −5 implies commuting is extremely sensitive to travel time; IV strengthens the magnitude, consistent with endogeneity bias in OLS.

TABLE I

GRAVITY EQUATION ESTIMATION USING 1921 BILATERAL COMMUTING DATA

	(1)	(2)
Second-stage regression		
	$\log \lambda_{nit}$	$\log \lambda_{nit}$
$\log d^{W}_{nit}$	−4.899***	−5.203***
	(0.062)	(0.069)
Workplace fixed effects	yes	yes
Residence fixed effects	yes	yes
Kleibergen-Paap (p-value)		.000
Estimation	OLS	IV
Observations	3,023	3,023
R-squared	0.851	—
First-stage regression		
		$\log d^{W}_{nit}$
$\log d^{S}_{ni}$		0.429***
		(0.003)
Workplace and residence fixed effects		yes
First-stage F-statistic		22,235
Observations		3,023
R-squared		0.949

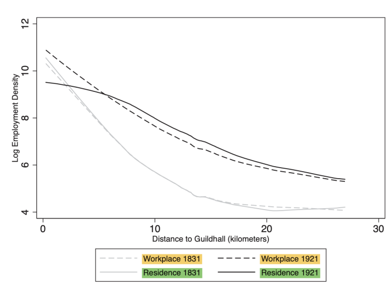


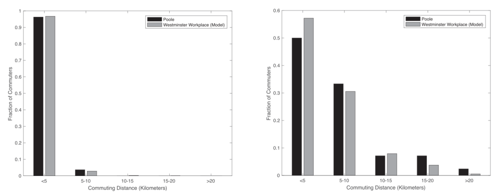
FIGURE VI

Employment Workplace and Residence Density and Distance to the Guildhall in 1831 and 1921

9.7 Figure VI: Employment density and residence density gradients

What it shows. Density profiles by distance to the Guildhall in 1831 vs 1921.

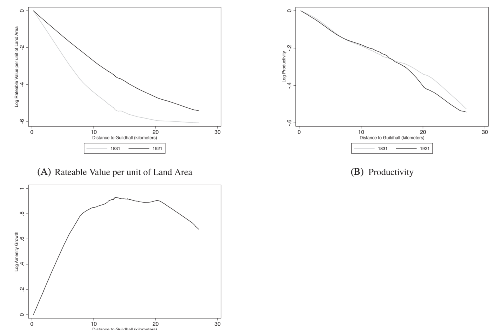
Interpretation. The city becomes more spatially structured: residence spreads outward (flatter or shifted residential gradient), while workplace density remains relatively central. This is the empirical signature of the "modern metropolis" pattern.



(A) 1857−1877 (B) 1891−1911

FIGURE VII

Commuting Distances in the Model and the Henry Poole Data



(A) Rateable Value per unit of Land Area (B) Productivity

(C) Amenity Growth 1831−1921

FIGURE VIII

Rateable Values and Productivity in 1831 and 1921 and Amenity Growth from 1831 to 1921

9.8 Figure VII: Commuting distance distribution (data vs model)

What it shows. Commuting distance distributions from the model and from Henry Poole's survey data (for 1900).

Interpretation. Matching the distance distribution is a strong validation target: it checks whether the model's commuting-cost mapping generates realistic long-distance commuting patterns.

9.9 Figure VIII: Rateable values and recovered productivity/amenity changes

What it shows. Panels relate rateable-value changes and recovered composite productivity/amenities to distance to center and time period.

Interpretation. Productivity shifts toward central areas and/or inner zones, while amenity changes capture residential desirability. These recovered fundamentals are what the counterfactuals hold fixed when changing the transport network.

9.10 Table II: Agglomeration forces

What it estimates. Elasticities of productivity with respect to workplace density (η^L) and amenities with respect to residential density (η^R), comparing OLS and IV.

Interpretation. Agglomeration is present and statistically significant; IV estimates are smaller but still economically meaningful, and they act as multipliers in network counterfactuals.

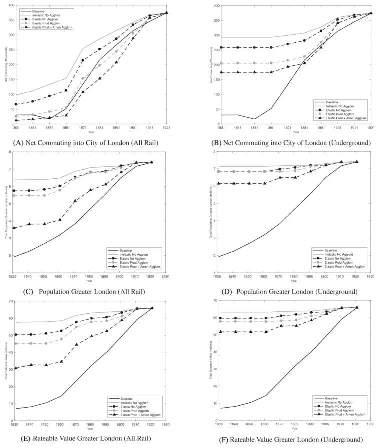
9.11 Figure IX: Counterfactual removal of the railway network

What it shows. Spatial consequences of removing the network (e.g. allocations, commuting, land values), under different assumptions.

TABLE II

ESTIMATION OF AGGLOMERATION FORCES IN PRODUCTION AND RESIDENCE

	$\ln \widehat{\mathbb{A}}^T_{nt}$ (1)	$\ln \widehat{\mathbb{B}}_{nt}$ (2)	$\ln \widehat{\mathbb{A}}^T_{nt}$ (3)	$\ln \widehat{\mathbb{B}}_{nt}$ (4)
$\ln \widehat{L}_{nt}$	0.148***	—	0.086**	—
	(0.027)		(0.037)	
$\ln \widehat{R}_{nt}$	—	0.248***	—	0.172***
		(0.023)		(0.031)
$\ln L_{nt}$	0.029*	—	0.011	—
	(0.017)		(0.017)	
$\ln R_{nt}$	—	0.033	—	−0.015
		(0.024)		(0.027)
$\ln K_n$	0.078***	−0.067	0.092***	−0.056
	(0.020)	(0.042)	(0.023)	(0.038)
$\mathbb{I}^{\text{LCC}}_n$	−0.112**	0.085	−0.033	0.252***
	(0.048)	(0.074)	(0.060)	0.089
First-stage F-statistic	—	—	11.26	12.76
Kleibergen-Paap (p-value)	—	—	.000	.000
Hansen-Sargan (p-value)	—	—	.416	.483
Estimation	OLS	OLS	IV	IV
Observations	99	99	99	99
R-squared	0.428	0.648	—	—



(A) Net Commuting into City of London (All Rail)
(B) Net Commuting into City of London (Underground)
(C) Population Greater London (All Rail)
(D) Population Greater London (Underground)
(E) Rateable Value Greater London (All Rail)
(F) Rateable Value Greater London (Underground)

FIGURE IX

Counterfactuals for Removing the Railway Network

TABLE III

COUNTERFACTUALS FOR REMOVING ALL OR PART OF THE RAILWAY NETWORK STARTING FROM THE INITIAL EQUILIBRIUM IN OUR BASELINE YEAR OF 1921

	(1)	(2)	(3)	(4)
Floor space supply elasticity	$\mu = 0$	$\mu = 1.83$	$\mu = 1.83$	$\mu = 1.83$
Production agglomeration force	$\eta^L = 0$	$\eta^L = 0$	$\eta^L = 0.086$	$\eta^L = 0.086$
Residential agglomeration force	$\eta^R = 0$	$\eta^R = 0$	$\eta^R = 0$	$\eta^R = 0.172$
Panel A: Removing the entire overground and underground railway network				
Economic impact				
Rateable value	−£8.24m	−£15.55m	−£20.78m	−£35.07m
NPV rateable value (3%)	−£274.55m	−£518.26m	−£692.76m	−£1,169.05m
NPV rateable value (5%)	−£164.73m	−£310.96m	−£415.66m	−£701.43m
Construction costs				
Cut-and-cover underground		−£9.96m		
Bored-tube underground		−£22.90m		
Overground railway		−£33.19m		
Total all railways		−£66.05m		
Ratio economic impact to construction cost				
$\frac{\text{NPV rateable value (3\%)}}{\text{Construction cost}}$	4.16	7.85	10.49	17.70
$\frac{\text{NPV rateable value (5\%)}}{\text{Construction cost}}$	2.49	4.71	6.29	10.62
Panel B: Removing the entire underground railway network				
Economic impact				
Rateable value	−£2.65m	−£6.21m	−£8.22m	−£14.16m
NPV rateable value (3%)	−£88.46m	−£206.87m	−£274.05m	−£471.85m
NPV rateable value (5%)	−£53.08m	−£124.12m	−£164.43m	−£283.11m
Construction costs				
Cut-and-cover underground		−£9.96m		
Bored-tube underground		−£22.90m		
Total all underground		−£32.86m		
Ratio economic impact to construction cost				
$\frac{\text{NPV rateable value (3\%)}}{\text{Construction cost}}$	2.69	6.30	8.34	14.36
$\frac{\text{NPV rateable value (5\%)}}{\text{Construction cost}}$	1.62	3.78	5.00	8.62
Panel C: Removing overground and underground railway lines constructed from 1911 to 1921				
Economic impact				
Rateable value	−£0.17m	−£0.24m	−£0.37m	−£0.39m
NPV rateable value (3%)	−£5.63m	−£8.09m	−£12.46m	−£12.96m
NPV rateable value (5%)	−£3.38m	−£4.86m	−£7.47m	−£7.77m
Construction costs				
Cut-and-cover underground		−£0.00m		
Bored-tube underground		−£2.35m		
Overground railway		−£0.34m		
Total all railways		−£2.69m		
Ratio economic impact to construction cost				
$\frac{\text{NPV rateable value (3\%)}}{\text{Construction cost}}$	2.09	3.01	4.63	4.82
$\frac{\text{NPV rateable value (5\%)}}{\text{Construction cost}}$	1.26	1.81	2.78	2.89

Interpretation. Railways are not just "marginal" infrastructure: they can rationalize a large part of the metropolis's spatial reorganization and implied welfare gains.

9.12 Table III: Welfare/value counterfactuals

What it reports. Rateable-value losses, NPVs, construction costs, and benefit/cost ratios for removing the network (all lines; underground only; recent lines).

Interpretation. The implied benefit/cost ratios are large in most specifications. Allowing for agglomeration and elastic floorspace supply substantially increases total implied gains.

10 Conclusion

10.1 Summary

Heblich, Redding, and Sturm (2020) show that the invention and expansion of the **steam railway** fundamentally reorganized London into a "modern metropolis" by enabling the first large-scale **separation of workplace and residence**. Using newly constructed, highly disaggregated spatial data for London from 1801 to 1921, the paper documents that central London (the City) experiences sharp **resident population decline** while simultaneously becoming more intensively used for **employment** and retaining high **land and building values** (captured by rateable values). To interpret these patterns, the authors develop a **structural estimation procedure** for a broad class of quantitative urban models that imply (i) a gravity structure in commuting flows and (ii) a constant proportional relationship between labor income and payments for floor space. Even though bilateral commuting flows are observed only in 1921, the framework leverages observed population and rateable values to infer how commuting costs and equilibrium allocations must have evolved throughout the nineteenth century. A canonical model within this class matches both the sharp divergence between day and night population in the City and the historical fact that, before the railway age, most people lived close to where they worked. Extending the model to allow for endogenous floor-space supply and agglomeration forces, the paper estimates economically meaningful agglomeration in both production and residence and shows that transport improvements can generate very large welfare/value gains.

10.2 Intuition: why railways create the "workplace center + residential suburbs"

The key mechanism is a commuting-technology shock with an inherently **hub-and-spoke** geometry:

1. **Railways disproportionately reduce commuting costs into central locations.** The network lowers travel times especially along radial routes to the historical center.

2. **If the center has high productivity but relatively weak residential amenities**, then cheaper commuting makes it optimal for the center to *specialize as a workplace* while outlying areas *specialize as residences*. Workers can live farther away (higher amenity or cheaper space) while still accessing high-wage central jobs.

3. **This specialization is reinforced by general equilibrium land markets.** As accessibility improves, willingness-to-pay for floor space re-allocates across space, and these changes are capitalized into local land/building values.

4. **Agglomeration amplifies the reorganization.** If productivity rises with workplace density (production agglomeration) and amenities rise with residential density (residential agglomeration), then concentrating employment in the center and residents in the suburbs further increases the center–suburb gap, magnifying the overall impact of the commuting shock.

10.3 Main empirical conclusions (reduced-form evidence)

The reduced-form evidence supports the mechanism in a tight sequence:

- After railway construction, **population falls in the City and rises in the suburbs**.
- The City's **population decline is combined with rising employment**, implying a switch from residential to workplace specialization.
- The City becomes **relatively more valuable** as a location (rateable values), consistent with intensified commercial use.
- Event-study evidence shows **no systematic deviations from parish trends before** station arrival and **large deviations after** arrival, with **negative relative effects in central areas** and **positive relative effects in suburban areas**.

10.4 Quantitative conclusions (structural estimates and counterfactuals)

Agglomeration forces. Under the identifying assumption that London's reorganization is driven by commuting-cost changes (and agglomeration) rather than systematic changes in locational fundamentals, the paper estimates substantial agglomeration in both production and residence. In the baseline IV estimates (Table II), the implied elasticities are approximately

$$\eta^L \approx 0.086 \qquad \text{(production agglomeration)}, \qquad \eta^R \approx 0.172 \qquad \text{(residential agglomeration)}.$$

Counterfactual: removing the railway network. The counterfactual exercises imply that **commuting-cost changes alone explain most of the observed separation of workplace and residence**. Holding floor-space supply, productivity, and amenities fixed, removing the railway network:

- reduces Greater London's **total population by about 13.7%**,
- reduces the **rateable value** of Greater London by about 12.5%, and
- decreases **net commuting into the City of London** by **more than 270,000 workers**.

Allowing for an **endogenous floor-space supply** and **agglomeration forces** substantially magnifies the impacts: using the calibrated floor-space supply elasticity and estimated agglomeration parameters, the declines in **population** and **rateable values** reach about

$$51.5\% \quad \text{and} \quad 53.3\%,$$

respectively.

Cost–benefit implication. Across a wide range of specifications, the implied increases in the **net present value of land and buildings** generated by the railway network exceed historical estimates of its construction costs (Table III), indicating very large economic payoffs to the transport investment once general equilibrium and amplification channels are accounted for.

10.5 Takeaway

By sharply lowering commuting costs into the center, the railway enabled London to separate residence from workplace, turning the historical core into an employment hub and the periphery into residential suburbs; quantitative urban models with commuting gravity and land-market clearing can explain this reorganization and imply very large welfare/value gains, especially when floor-space supply and agglomeration are allowed to respond endogenously.